Resource-Efficient Evidence Access: Kaggle Evaluation

Draft evidence status: complete combined-report members were read from an incomplete archive download. All displayed counts reconcile with the saved metrics, but full raw-database validation remains pending. This T4/FP16 run is separate from the bounded-interface study in the existing manuscript.

On 622 questions scored by both SemPointer and full context, SemPointer used 76.7% fewer active input tokens and 61.3% lower reported per-query latency. Across the same 2,400 questions as matched-budget BM25, it used essentially the same mean active input budget and recorded 8.9% lower reported latency. These measurements characterize an efficient evidence-access operating point for the evaluated checkpoint and hardware.

Evaluation setting

This supplement reports frozen run 8ef26bfd68509bae3b6b using Qwen3.5-4B (revision 851bf6e806efd8d0a36b00ddf55e13ccb7b8cd0a), NF4 weights, FP16 compute, batch size one, and one independent worker on each of two NVIDIA T4 GPUs. Eight 300-question dataset subsets are complete: BABILong, InfiniteBench, L-Eval, LoCoMo, LongBench, LongBench v2, LongMemEval, and LooGLE. Their first five systems score the same 2,400 selected questions. The archived snapshot records 13,055 scored outcomes and 1,798 skips across all ten selected families; RULER and SCROLLS have partial coverage. The planned 3,000 questions produce up to 18,000 system-question outcomes, rather than 18,000 independent questions.

Active tokens count the model input reported by the frozen evaluator. Reported latency includes generation and measured selector cost; shared ingestion and outer prompt preparation are accounted for separately. Means below are question-weighted, and savings are ratios of paired totals. Both workers operate concurrently, but the latency column measures individual requests rather than serving throughput. Timing percentages describe this recorded execution; no repeat-run timing interval has been estimated.

Resource measurements

Within the 622-question full-context-eligible subset, mean active input falls from 8,038.8 to 1,875.1 tokens. Mean reported latency falls from 12.564 to 4.865 s, corresponding to 76.7% and 61.3% reductions, respectively. Eligibility is defined by the original evaluator and its 16,384-token prompt limit; these resource estimates apply to the scored shared subset.

Over all 2,400 questions in the eight complete subsets, SemPointer and matched-budget BM25 use 4,071.4 and 4,071.3 active input tokens on average. Reported latency is 5.504 versus 6.045 s, a 8.9% reduction at an effectively matched input budget.

Comparison	Paired n	System	Active tokens	Latency (s)
Matched retrieval budget	2400	SemPointer	4,071.4	5.504
Matched retrieval budget	2400	Matched-budget BM25	4,071.3	6.045
Eligible full context	622	SemPointer	1,875.1	4.865
Eligible full context	622	Full context	8,038.8	12.564

Table 1. Question-weighted resource measurements. Comparisons are paired within each block; active tokens and measured per-query latency retain the frozen evaluator definitions.

Illustrative matched-budget quality results

Table 2 presents illustrative matched-budget operating points selected after inspection of the recorded task inventory. On LoCoMo QA, SemPointer scores 24.73 versus 24.69 with approximately 2,197 active tokens in each system. On HotpotQA it records 73.28 versus 68.13; on RepoBench-P it records 24.69 versus 20.94. InfiniteBench key-value retrieval scores 80.00 in both systems. These examples span conversational memory, multi-hop QA, code completion, and retrieval. Each comparison uses the same scored question IDs and the task-specific adapted metric.

Task	Paired n	SemPointer	Matched BM25	Difference	Paired 95% interval
LoCoMo QA	300	24.73	24.69	+0.04	[-2.52, +2.51]
LongBench HotpotQA	27	73.28	68.13	+5.15	[-5.65, +17.59]
LongBench Qasper	8	50.15	39.48	+10.68	[+0.00, +31.78]
LongBench RepoBench-P	26	24.69	20.94	+3.75	[-1.54, +11.10]
InfiniteBench KV retrieval	25	80.00	80.00	+0.00	[+0.00, +0.00]

Table 2. Adapted F1 for LoCoMo, HotpotQA, and Qasper; code similarity for RepoBench-P; containment for KV retrieval. Scores and differences use a 0-100 display scale. Examples were selected after inspection.

Quality scores are shown on a 0-100 display scale while retaining each task metric. Confidence intervals are the frozen evaluator's paired example-level bootstrap intervals, with no multiple-comparison adjustment. Shared source documents and conversations can introduce dependence between questions. Missing scores and recorded skips are distinguished from zero scores. These are adapted local benchmark metrics; code similarity is not executable-code correctness. The selected examples are illustrative rather than a prespecified confirmatory test. No heterogeneous overall accuracy is computed.

Empirical interpretation

Together, the measurements support lower active-input and recorded request cost relative to eligible full context, and lower recorded latency at a matched retrieval budget. The quality examples document task-specific operating points; their score intervals are reported directly, without an equivalence or universal-improvement claim. The complete all-system inventory accompanies these examples in Appendix A.

Appendix A. Complete all-system score inventory

All 138 selected task groups are retained, including five groups without a recorded score. Each score is followed by its scored n in parentheses. A dash means no scored result; it does not mean zero. Skip and unrecorded counts are retained in the CSV files. Full-context cells often describe smaller subsets.

Dataset	Task	Metric	Selected n	SemPointer	Pointer off	BM25-2	BM25-4	Matched BM25	Full context
babilong	data/qa1/128k	exact	7	28.57 (7)	14.29 (7)	14.29 (7)	14.29 (7)	0.00 (7)	-
babilong	data/qa1/16k	exact	3	0.00 (3)	0.00 (3)	0.00 (3)	0.00 (3)	0.00 (3)	0.00 (3)
babilong	data/qa1/256k	exact	3	33.33 (3)	33.33 (3)	33.33 (3)	33.33 (3)	33.33 (3)	-
babilong	data/qa1/32k	exact	5	20.00 (5)	20.00 (5)	20.00 (5)	0.00 (5)	0.00 (5)	-
babilong	data/qa1/4k	exact	2	50.00 (2)	0.00 (2)	0.00 (2)	0.00 (2)	0.00 (2)	0.00 (2)
babilong	data/qa1/64k	exact	5	20.00 (5)	0.00 (5)	0.00 (5)	0.00 (5)	0.00 (5)	-
babilong	data/qa1/8k	exact	5	40.00 (5)	40.00 (5)	40.00 (5)	20.00 (5)	20.00 (5)	20.00 (5)
babilong	data/qa10/128k	exact	5	40.00 (5)	60.00 (5)	60.00 (5)	80.00 (5)	20.00 (5)	-
babilong	data/qa10/16k	exact	3	33.33 (3)	33.33 (3)	33.33 (3)	66.67 (3)	66.67 (3)	33.33 (3)
babilong	data/qa10/256k	exact	5	60.00 (5)	40.00 (5)	40.00 (5)	80.00 (5)	60.00 (5)	-
babilong	data/qa10/32k	exact	4	75.00 (4)	75.00 (4)	50.00 (4)	75.00 (4)	50.00 (4)	-
babilong	data/qa10/4k	exact	4	50.00 (4)	50.00 (4)	50.00 (4)	50.00 (4)	75.00 (4)	50.00 (4)
babilong	data/qa10/64k	exact	6	33.33 (6)	33.33 (6)	33.33 (6)	33.33 (6)	50.00 (6)	-
babilong	data/qa10/8k	exact	3	100.00 (3)	100.00 (3)	100.00 (3)	100.00 (3)	100.00 (3)	100.00 (3)
babilong	data/qa2/128k	exact	6	0.00 (6)	0.00 (6)	0.00 (6)	0.00 (6)	0.00 (6)	-
babilong	data/qa2/16k	exact	2	0.00 (2)	0.00 (2)	0.00 (2)	0.00 (2)	0.00 (2)	0.00 (2)
babilong	data/qa2/256k	exact	4	0.00 (4)	0.00 (4)	0.00 (4)	0.00 (4)	0.00 (4)	-
babilong	data/qa2/32k	exact	6	16.67 (6)	16.67 (6)	16.67 (6)	16.67 (6)	16.67 (6)	-
babilong	data/qa2/64k	exact	4	0.00 (4)	0.00 (4)	0.00 (4)	0.00 (4)	0.00 (4)	-
babilong	data/qa2/8k	exact	8	0.00 (8)	0.00 (8)	0.00 (8)	0.00 (8)	0.00 (8)	12.50 (8)
babilong	data/qa3/128k	exact	6	0.00 (6)	0.00 (6)	0.00 (6)	0.00 (6)	0.00 (6)	-
babilong	data/qa3/16k	exact	4	0.00 (4)	0.00 (4)	0.00 (4)	0.00 (4)	0.00 (4)	0.00 (4)
babilong	data/qa3/256k	exact	4	0.00 (4)	0.00 (4)	0.00 (4)	0.00 (4)	25.00 (4)	-
babilong	data/qa3/32k	exact	6	16.67 (6)	16.67 (6)	16.67 (6)	33.33 (6)	16.67 (6)	-
babilong	data/qa3/4k	exact	4	0.00 (4)	0.00 (4)	0.00 (4)	50.00 (4)	0.00 (4)	25.00 (4)
babilong	data/qa3/64k	exact	4	0.00 (4)	0.00 (4)	0.00 (4)	0.00 (4)	0.00 (4)	-

Dataset	Task	Metric	Selected n	SemPointer	Pointer off	BM25-2	BM25-4	Matched BM25	Full context
babilong	data/qa3/8k	exact	2	0.00 (2)	0.00 (2)	0.00 (2)	0.00 (2)	0.00 (2)	50.00 (2)
babilong	data/qa4/128k	exact	3	100.00 (3)	100.00 (3)	100.00 (3)	100.00 (3)	100.00 (3)	-
babilong	data/qa4/16k	exact	4	75.00 (4)	75.00 (4)	75.00 (4)	100.00 (4)	75.00 (4)	66.67 (3)
babilong	data/qa4/256k	exact	7	57.14 (7)	57.14 (7)	57.14 (7)	57.14 (7)	57.14 (7)	-
babilong	data/qa4/32k	exact	8	62.50 (8)	50.00 (8)	50.00 (8)	50.00 (8)	50.00 (8)	-
babilong	data/qa4/4k	exact	4	25.00 (4)	25.00 (4)	25.00 (4)	25.00 (4)	25.00 (4)	25.00 (4)
babilong	data/qa4/64k	exact	2	100.00 (2)	100.00 (2)	100.00 (2)	100.00 (2)	100.00 (2)	-
babilong	data/qa4/8k	exact	2	100.00 (2)	100.00 (2)	100.00 (2)	100.00 (2)	100.00 (2)	100.00 (2)
babilong	data/qa5/128k	exact	6	50.00 (6)	33.33 (6)	33.33 (6)	33.33 (6)	33.33 (6)	-
babilong	data/qa5/16k	exact	2	50.00 (2)	50.00 (2)	50.00 (2)	50.00 (2)	50.00 (2)	50.00 (2)
babilong	data/qa5/256k	exact	4	25.00 (4)	50.00 (4)	50.00 (4)	75.00 (4)	100.00 (4)	-
babilong	data/qa5/32k	exact	5	60.00 (5)	40.00 (5)	40.00 (5)	60.00 (5)	60.00 (5)	-
babilong	data/qa5/4k	exact	1	0.00 (1)	0.00 (1)	0.00 (1)	0.00 (1)	100.00 (1)	100.00 (1)
babilong	data/qa5/64k	exact	5	60.00 (5)	60.00 (5)	60.00 (5)	40.00 (5)	40.00 (5)	-
babilong	data/qa5/8k	exact	7	57.14 (7)	57.14 (7)	57.14 (7)	57.14 (7)	57.14 (7)	42.86 (7)
babilong	data/qa6/128k	exact	7	85.71 (7)	85.71 (7)	85.71 (7)	71.43 (7)	57.14 (7)	-
babilong	data/qa6/16k	exact	3	66.67 (3)	100.00 (3)	100.00 (3)	66.67 (3)	33.33 (3)	0.00 (3)
babilong	data/qa6/256k	exact	3	100.00 (3)	66.67 (3)	66.67 (3)	100.00 (3)	66.67 (3)	-
babilong	data/qa6/32k	exact	5	100.00 (5)	100.00 (5)	100.00 (5)	100.00 (5)	80.00 (5)	-
babilong	data/qa6/4k	exact	4	100.00 (4)	75.00 (4)	75.00 (4)	75.00 (4)	100.00 (4)	100.00 (4)
babilong	data/qa6/64k	exact	5	60.00 (5)	60.00 (5)	60.00 (5)	60.00 (5)	60.00 (5)	-
babilong	data/qa6/8k	exact	3	100.00 (3)	33.33 (3)	33.33 (3)	66.67 (3)	100.00 (3)	100.00 (3)
babilong	data/qa7/128k	exact	3	0.00 (3)	0.00 (3)	0.00 (3)	0.00 (3)	33.33 (3)	-
babilong	data/qa7/16k	exact	4	0.00 (4)	0.00 (4)	0.00 (4)	25.00 (4)	0.00 (4)	0.00 (4)
babilong	data/qa7/256k	exact	7	14.29 (7)	42.86 (7)	14.29 (7)	57.14 (7)	28.57 (7)	-
babilong	data/qa7/32k	exact	3	0.00 (3)	0.00 (3)	0.00 (3)	0.00 (3)	0.00 (3)	-
babilong	data/qa7/4k	exact	4	50.00 (4)	25.00 (4)	25.00 (4)	0.00 (4)	0.00 (4)	0.00 (4)
babilong	data/qa7/64k	exact	7	14.29 (7)	28.57 (7)	14.29 (7)	28.57 (7)	14.29 (7)	-
babilong	data/qa7/8k	exact	2	0.00 (2)	0.00 (2)	0.00 (2)	0.00 (2)	0.00 (2)	0.00 (2)
babilong	data/qa8/128k	exact	7	28.57 (7)	14.29 (7)	14.29 (7)	14.29 (7)	42.86 (7)	-

Dataset	Task	Metric	Selected n	SemPointer	Pointer off	BM25-2	BM25-4	Matched BM25	Full context
babilong	data/qa8/16k	exact	3	66.67 (3)	66.67 (3)	66.67 (3)	66.67 (3)	33.33 (3)	33.33 (3)
babilong	data/qa8/256k	exact	3	0.00 (3)	0.00 (3)	33.33 (3)	0.00 (3)	0.00 (3)	-
babilong	data/qa8/32k	exact	2	50.00 (2)	50.00 (2)	50.00 (2)	50.00 (2)	50.00 (2)	-
babilong	data/qa8/4k	exact	2	50.00 (2)	50.00 (2)	50.00 (2)	50.00 (2)	50.00 (2)	50.00 (2)
babilong	data/qa8/64k	exact	8	25.00 (8)	25.00 (8)	25.00 (8)	25.00 (8)	25.00 (8)	-
babilong	data/qa8/8k	exact	5	0.00 (5)	0.00 (5)	0.00 (5)	20.00 (5)	20.00 (5)	20.00 (5)
babilong	data/qa9/128k	exact	3	100.00 (3)	100.00 (3)	100.00 (3)	100.00 (3)	100.00 (3)	-
babilong	data/qa9/16k	exact	2	100.00 (2)	50.00 (2)	50.00 (2)	50.00 (2)	50.00 (2)	50.00 (2)
babilong	data/qa9/256k	exact	7	42.86 (7)	71.43 (7)	57.14 (7)	100.00 (7)	28.57 (7)	-
babilong	data/qa9/32k	exact	5	80.00 (5)	60.00 (5)	80.00 (5)	80.00 (5)	80.00 (5)	-
babilong	data/qa9/4k	exact	4	75.00 (4)	75.00 (4)	75.00 (4)	100.00 (4)	75.00 (4)	100.00 (4)
babilong	data/qa9/64k	exact	5	100.00 (5)	100.00 (5)	100.00 (5)	100.00 (5)	100.00 (5)	-
babilong	data/qa9/8k	exact	4	100.00 (4)	100.00 (4)	100.00 (4)	100.00 (4)	100.00 (4)	50.00 (4)
infinite	code_debug	choice	26	34.62 (26)	42.31 (26)	26.92 (26)	42.31 (26)	57.69 (26)	-
infinite	code_run	numeric_sequence	25	0.00 (25)	0.00 (25)	0.00 (25)	0.00 (25)	0.00 (25)	-
infinite	kv_retrieval	all_containment	25	80.00 (25)	76.00 (25)	76.00 (25)	76.00 (25)	80.00 (25)	-
infinite	longbook_choice_eng	choice	25	60.00 (25)	64.00 (25)	52.00 (25)	68.00 (25)	68.00 (25)	-
infinite	longbook_qa_chn	char_f1	24	11.15 (24)	11.85 (24)	8.79 (24)	10.93 (24)	10.93 (24)	-
infinite	longbook_qa_eng	qa_f1	25	23.81 (25)	21.57 (25)	20.92 (25)	32.84 (25)	31.73 (25)	-
infinite	longbook_sum_eng	rouge_l	25	2.46 (25)	1.77 (25)	1.41 (25)	2.19 (25)	3.37 (25)	-
infinite	longdialogue_qa_eng	qa_f1	25	16.00 (25)	18.00 (25)	18.00 (25)	20.00 (25)	20.00 (25)	-
infinite	math_calc	numeric_sequence	25	0.00 (25)	0.00 (25)	0.00 (25)	0.00 (25)	0.00 (25)	-
infinite	math_find	numeric_sequence	25	32.00 (25)	40.00 (25)	36.00 (25)	40.00 (25)	40.00 (25)	-
infinite	number_string	all_containment	25	100.00 (25)	100.00 (25)	100.00 (25)	100.00 (25)	100.00 (25)	-
infinite	passkey	all_containment	25	100.00 (25)	100.00 (25)	100.00 (25)	100.00 (25)	100.00 (25)	-
leval	codeU	numeric_sequence	17	0.00 (17)	0.00 (17)	0.00 (17)	0.00 (17)	5.88 (17)	-
leval	coursera	choice_set	17	58.82 (17)	58.82 (17)	64.71 (17)	64.71 (17)	58.82 (17)	58.82 (17)
leval	financial_qa	qa_f1	17	40.24 (17)	41.51 (17)	39.88 (17)	47.98 (17)	43.56 (17)	47.57 (17)
leval	gov_report_summ	rouge_l	11	14.21 (11)	14.25 (11)	14.06 (11)	15.37 (11)	13.98 (11)	15.94 (10)
leval	gsm100	numeric_sequence	17	0.00 (17)	0.00 (17)	0.00 (17)	0.00 (17)	0.00 (17)	0.00 (17)

Dataset	Task	Metric	Selected n	SemPointer	Pointer off	BM25-2	BM25-4	Matched BM25	Full context
leval	legal_contract_qa	qa_f1	17	33.17 (17)	36.41 (17)	24.05 (17)	35.80 (17)	34.14 (17)	41.64 (9)
leval	meeting_summ	rouge_l	16	15.80 (16)	16.56 (16)	17.50 (16)	16.34 (16)	17.85 (16)	16.98 (8)
leval	multidoc_qa	qa_f1	16	33.36 (16)	31.20 (16)	29.17 (16)	41.24 (16)	36.96 (16)	42.89 (16)
leval	narrative_qa	qa_f1	16	13.17 (16)	8.48 (16)	5.03 (16)	16.28 (16)	19.69 (16)	23.53 (2)
leval	natural_question	qa_f1	16	51.88 (16)	49.29 (16)	49.29 (16)	63.78 (16)	58.57 (16)	61.52 (8)
leval	news_summ	rouge_l	8	10.38 (8)	10.75 (8)	9.34 (8)	11.56 (8)	11.22 (8)	11.98 (8)
leval	paper_assistant	rouge_l	16	15.51 (16)	15.16 (16)	14.82 (16)	16.34 (16)	15.12 (16)	17.41 (15)
leval	patent_summ	rouge_l	9	23.48 (9)	23.20 (9)	22.78 (9)	26.21 (9)	24.22 (9)	25.78 (9)
leval	quality	choice	16	87.50 (16)	93.75 (16)	75.00 (16)	100.00 (16)	93.75 (16)	100.00 (16)
leval	review_summ	rouge_l	16	13.81 (16)	12.43 (16)	11.43 (16)	13.92 (16)	14.49 (16)	-
leval	sci_fi	exact	16	81.25 (16)	68.75 (16)	50.00 (16)	68.75 (16)	75.00 (16)	100.00 (9)
leval	scientific_qa	qa_f1	16	28.37 (16)	25.77 (16)	25.77 (16)	39.01 (16)	31.27 (16)	45.05 (16)
leval	topic_retrieval_longchat	qa_f1	16	24.34 (16)	23.05 (16)	23.05 (16)	29.95 (16)	20.67 (16)	95.67 (16)
leval	tpo	choice	16	87.50 (16)	87.50 (16)	87.50 (16)	93.75 (16)	87.50 (16)	93.75 (16)
leval	tv_show_summ	rouge_l	11	11.49 (11)	12.43 (11)	12.16 (11)	14.19 (11)	12.78 (11)	14.30 (11)
locomo	qa	qa_f1	300	24.73 (300)	23.49 (300)	22.36 (300)	25.33 (300)	24.69 (300)	-
longbench	hotpotqa	qa_f1	27	73.28 (27)	62.02 (27)	54.33 (27)	68.19 (27)	68.13 (27)	77.22 (15)
longbench	lcc	code_similarity	24	14.89 (24)	13.83 (24)	16.49 (24)	13.83 (24)	14.17 (24)	14.48 (23)
longbench	lsht	exact	13	38.46 (13)	38.46 (13)	53.85 (13)	38.46 (13)	38.46 (13)	62.50 (8)
longbench	multi_news	rouge_l	17	5.46 (17)	5.57 (17)	5.42 (17)	5.57 (17)	6.14 (17)	8.99 (17)
longbench	multifieldqa_en	qa_f1	8	50.74 (8)	52.82 (8)	55.72 (8)	51.89 (8)	50.51 (8)	47.05 (8)
longbench	multifieldqa_zh	char_f1	34	49.54 (34)	53.06 (34)	45.69 (34)	55.14 (34)	50.51 (34)	63.57 (34)
longbench	musique	qa_f1	23	26.74 (23)	23.63 (23)	25.60 (23)	25.66 (23)	25.53 (23)	37.04 (6)
longbench	passage_count	numeric_sequence	13	0.00 (13)	0.00 (13)	0.00 (13)	0.00 (13)	0.00 (13)	0.00 (8)
longbench	passage_retrieval_en	all_containment	11	0.00 (11)	0.00 (11)	0.00 (11)	0.00 (11)	0.00 (11)	0.00 (11)
longbench	passage_retrieval_zh	all_containment	26	7.69 (26)	0.00 (26)	0.00 (26)	0.00 (26)	0.00 (26)	3.85 (26)
longbench	qasper	qa_f1	8	50.15 (8)	38.32 (8)	25.82 (8)	37.07 (8)	39.48 (8)	34.36 (8)
longbench	qmsum	rouge_l	15	12.94 (15)	13.02 (15)	12.72 (15)	14.23 (15)	15.25 (15)	15.73 (8)
longbench	repobench-p	code_similarity	26	24.69 (26)	24.41 (26)	21.92 (26)	24.05 (26)	20.94 (26)	20.66 (16)
longbench	samsum	qa_f1	17	46.58 (17)	44.77 (17)	48.01 (17)	44.79 (17)	46.50 (17)	47.86 (16)

Dataset	Task	Metric	Selected n	SemPointer	Pointer off	BM25-2	BM25-4	Matched BM25	Full context
longbench	trec	exact	10	70.00 (10)	70.00 (10)	80.00 (10)	70.00 (10)	70.00 (10)	80.00 (10)
longbench	triviaqa	qa_f1	10	93.33 (10)	93.33 (10)	93.33 (10)	93.33 (10)	93.33 (10)	90.00 (6)
longbench	vcsum	char_rouge_l	18	9.62 (18)	13.36 (18)	10.98 (18)	13.36 (18)	11.36 (18)	15.90 (18)
longbench_v2	data	choice	300	31.00 (300)	29.00 (300)	33.00 (300)	28.67 (300)	33.00 (300)	46.15 (13)
longmem	longmemeval_s_cleaned	qa_f1	300	36.65 (300)	35.92 (300)	28.64 (300)	36.25 (300)	37.60 (300)	-
loogle	longdep_qa	qa_f1	75	17.92 (75)	17.47 (75)	15.02 (75)	18.59 (75)	20.08 (75)	21.67 (5)
loogle	shortdep_cloze	all_containment	75	52.93 (75)	51.78 (75)	45.71 (75)	53.78 (75)	51.87 (75)	-
loogle	shortdep_qa	qa_f1	75	59.70 (75)	56.41 (75)	56.46 (75)	60.19 (75)	61.32 (75)	57.47 (22)
loogle	summarization	rouge_l	75	16.51 (75)	16.65 (75)	15.43 (75)	17.69 (75)	18.56 (75)	19.82 (25)
ruler	cwe	all_containment	51	2.50 (48)	2.50 (48)	2.55 (47)	14.26 (47)	1.70 (47)	13.42 (38)
ruler	niah_multikey_1	all_containment	51	100.00 (22)	100.00 (22)	100.00 (22)	100.00 (22)	100.00 (22)	100.00 (15)
ruler	niah_single_1	not started	51	-	-	-	-	-	-
ruler	niah_single_2	not started	51	-	-	-	-	-	-
ruler	niah_single_3	all_containment	48	100.00 (1)	100.00 (1)	100.00 (1)	100.00 (1)	100.00 (1)	-
ruler	vt	all_containment	48	60.00 (1)	60.00 (1)	60.00 (1)	80.00 (1)	60.00 (1)	100.00 (1)
scrolls	contract_nli	exact	43	0.00 (1)	0.00 (1)	0.00 (1)	0.00 (1)	0.00 (1)	0.00 (1)
scrolls	gov_report	rouge_l	43	7.02 (1)	4.77 (1)	4.77 (1)	8.02 (1)	7.16 (1)	10.64 (1)
scrolls	narrative_qa	qa_f1	43	12.77 (1)	19.05 (1)	19.05 (1)	24.00 (1)	45.45 (1)	-
scrolls	qasper	not started	43	-	-	-	-	-	-
scrolls	qmsum	rouge_l	43	16.98 (1)	16.98 (1)	16.98 (1)	16.67 (1)	16.67 (1)	-
scrolls	quality	not started	43	-	-	-	-	-	-
scrolls	summ_screen_fd	not started	42	-	-	-	-	-	-

Appendix B. Coverage and resource accounting

Dataset	Selected questions	Scored outcomes	Skipped outcomes	Unrecorded outcomes	Status
babilong	300	1599	201	0	Complete
infinite	300	1500	300	0	Complete
leval	300	1720	80	0	Complete
locomo	300	1500	300	0	Complete
longbench	300	1738	62	0	Complete
longbench_v2	300	1513	287	0	Complete
longmem	300	1500	300	0	Complete
loogle	300	1552	248	0	Complete
ruler	300	411	18	1371	Partial
scrolls	300	22	2	1776	Partial

Resource averages. The first five systems score the same 2,400 questions. Full context scores 622; its raw means below describe that eligible subset. Table 1 supplies the matched full-context comparison.

System	Scored n	Active tokens	Latency (s)	Generation (s)	TTFT (s)	Peak memory*
SemPointer	2400	4071.4	5.504	5.497	3.618	4168.1
Pointer off	2400	1739.4	3.531	3.524	1.535	3632.0
BM25-2	2400	1159.8	2.892	2.891	1.075	3498.3
BM25-4	2400	2190.8	3.951	3.950	1.881	3735.0
Matched-budget BM25	2400	4071.3	6.045	5.842	3.614	4168.1
Full context	622	8038.8	12.564	12.564	7.591	5080.8

* Memory retains the source report's MB units and is the mean of recorded per-question peak measurements. CSV files preserve all 828 system/task cells and all 605 paired task/control comparisons. Source metrics SHA256: 8ff2dde050f242a8de09434d270f10bc3a0d989aa92f3cea02fab8e17aaedf74